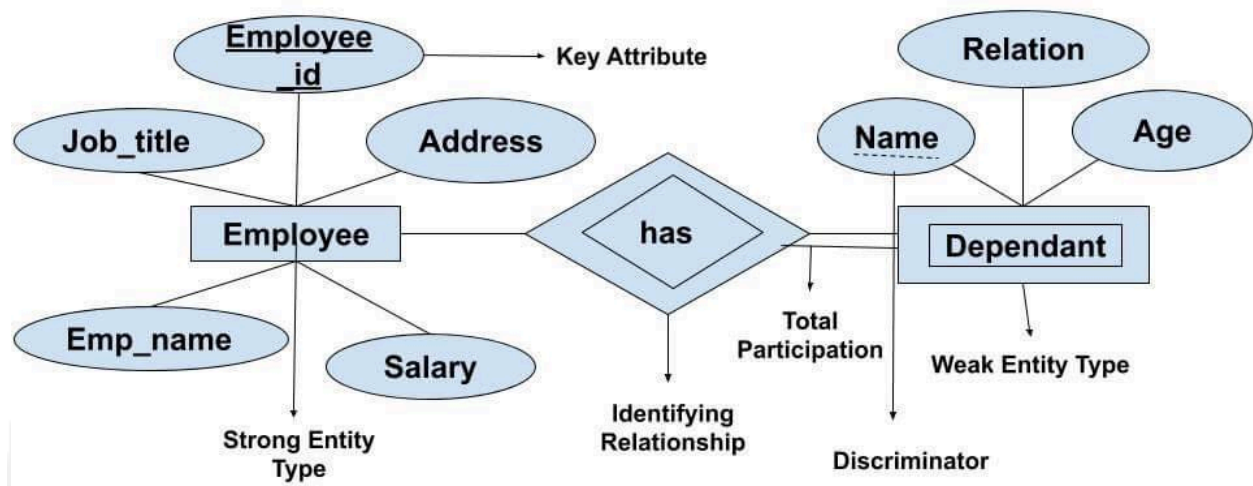
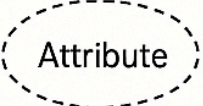
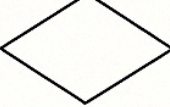
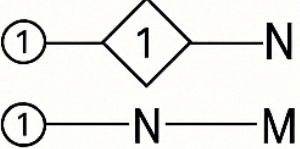
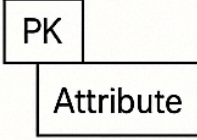
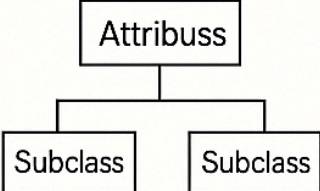


Comprehensive Analysis of the Entity-Relationship (ER) Model in DBMS



Multivalued Attribute  An attribute that can have multiple values	Derived Attribute  An attribute that is derived from another attribute	Identifying Relationship  A relationship that links a weak entity to its owner
Cardinality  The number of instances	Primary Key  A unique identifier	Foreign Key 

Executive Summary

The **Entity-Relationship (ER) Model** is the **conceptual foundation** of database design. It precedes table creation and SQL implementation and functions much like an **architectural blueprint**—ensuring that data is structured logically *before* it is stored physically.

Key conclusions:

- **ER modeling is mandatory planning**, not optional documentation.
- **Keys form a strict hierarchy**, enabling uniqueness, identification, and referential integrity.
- **ER diagrams use standardized symbols**, ensuring a universal design language.
- **Referential integrity constraints** govern safe insertion, deletion, and updates.
- **ER → RDBMS conversion** follows deterministic mapping rules tested heavily in exams.

1. Conceptual Framework of the ER Model

The ER model is a **high-level conceptual data model**, independent of:

- Storage
- File organization
- Indexing
- SQL syntax

It answers one core question:

What data exists, and how is it logically related?

Design Workflow

1. **ER Modeling**
 - Identify entities, attributes, relationships, and constraints
2. **RDBMS Conversion**
 - Convert ER structures into relational tables
3. **Refinement**
 - Apply normalization, constraints, and transaction control

Exam Insight

ER modeling belongs to the **design phase**, not the implementation phase.

2. Database Keys and Identification

Keys ensure **uniqueness, identification, and relationship enforcement**.

Key Hierarchy (Most Important Concept)

Key Type	Description	Key Properties
Super Key	Any attribute set that uniquely identifies a tuple	May contain extra attributes
Candidate Key	Minimal super key	No redundant attributes
Primary Key	Chosen candidate key	Unique + NOT NULL
Alternate Key	Candidate keys not chosen as PK	Still unique
Foreign Key	Attribute referencing another table's PK	Can be NULL and duplicate
Composite Key	Key with ≥ 2 attributes	Common in weak entities

Golden Rule

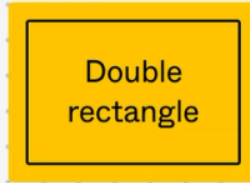
Primary Key $\subset$ Candidate Key $\subset$ Super Key

3. ER Model Components and Visualization

ER DIAGRAM SYMBOLS



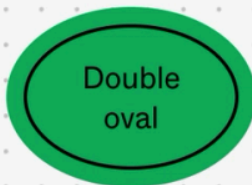
Strong entity



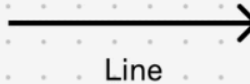
Weak entity



Attribute



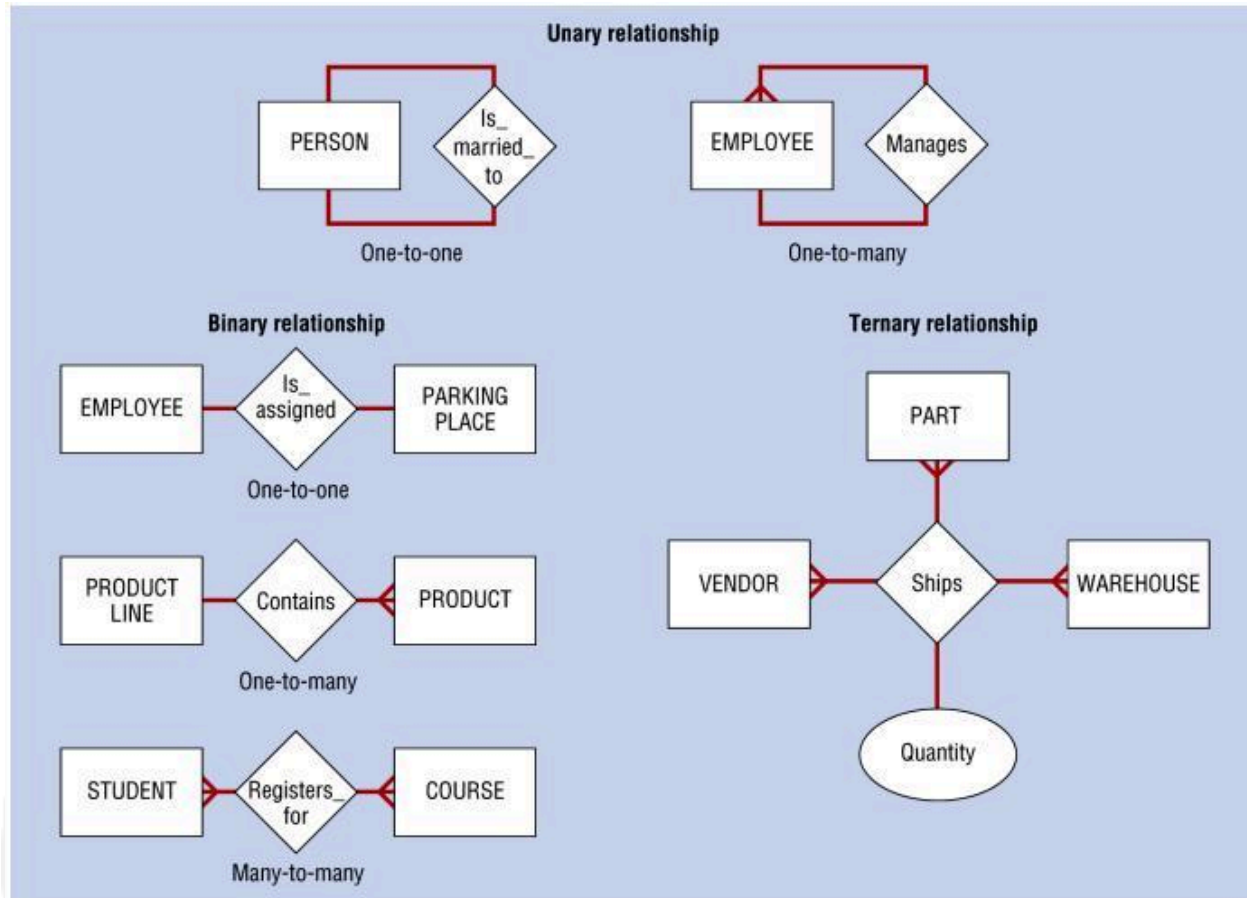
Mutli-value
attribute



Relationship or
connection



Description of
relationship



3.1 Entities and Entity Sets

- **Entity:** Real-world object (Student #101)
- **Entity Set:** Collection of similar entities (All Students)

Types

- **Strong Entity**
 - Exists independently
 - Rectangle
- **Weak Entity**
 - Depends on strong entity
 - Double rectangle
 - Uses **partial key**
 - Requires **identifying relationship**

3.2 Relationships

Relationships define **associations** among entities.

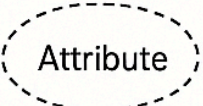
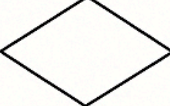
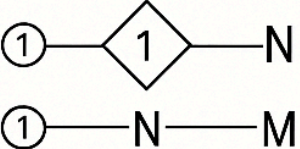
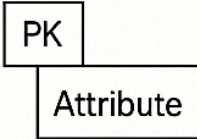
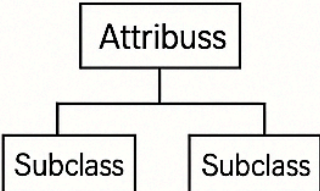
- Represented by **Diamonds**
- **Relationship Set**: Collection of same-type relationships

Degree of Relationships

- **Unary**: Entity relates to itself
- **Binary**: Most common (Student–Course)
- **Ternary**: Three entity sets
- **N-ary**: More than three

Binary $\neq$ always correct $\rightarrow$ ternary cannot always be decomposed safely.

3.3 Attributes

Multivalued Attribute  An attribute that can have multiple values	Derived Attribute  An attribute that is derived from another attribute	Identifying Relationship  A relationship that links a weak entity to its owner
Cardinality  The number of instances	Primary Key  A unique identifier	Foreign Key 

Attribute Type	Representation	Meaning
Key	Underlined oval	Uniquely identifies entity

Simple	Single oval	Atomic
Composite	Oval with branches	Divisible (Address)
Multi-valued	Double oval	Multiple values
Derived	Dashed oval	Computed (Age)

Derived attributes are **not stored** in tables.

4. Constraints in the ER Model

Constraints enforce **semantic correctness**.

4.1 Mapping Cardinalities

Defines **how many entities participate** in a relationship.

Type	Meaning
1:1	One ↔ One
1:N	One ↔ Many
N:1	Many ↔ One
M:N	Many ↔ Many

Arrows point toward the “**one**” side.

4.2 Participation Constraints

Defines **mandatory vs optional participation**.

- **Total Participation**
 - Every entity must participate
 - Double line
- **Partial Participation**
 - Optional participation

- Single line

Weak entities always have **total participation**.

5. Referential Integrity and Operations

artist_id	artist_name
1	Bono
2	Cher
3	Nuno Bettencourt

Link Broken



artist_id	album_id	album_name
3	1	Schizophonic
4	2	Eat the rich
3	3	Crave (single)

```

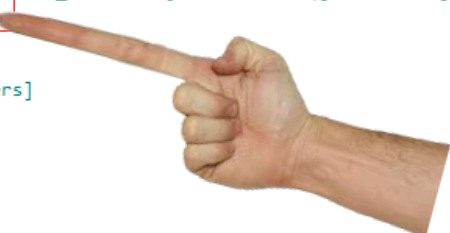
SQLQuery10.sql - L...RRE\mgledhill (52))* x SQLQuery9.sql - us...RRE\mgledhill (60))
USE [Northwind]
GO

ALTER TABLE [dbo].[Orders] DROP CONSTRAINT [FK_Orders_Customers]
GO

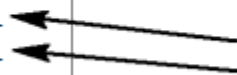
ALTER TABLE [dbo].[Orders] WITH NOCHECK ADD CONSTRAINT [FK_Orders_Customers] FOREIGN KEY([CustomerID])
REFERENCES [dbo].[Customers] ([CustomerID]) ON DELETE CASCADE
GO

ALTER TABLE [dbo].[Orders] CHECK CONSTRAINT [FK_Orders_Customers]
GO

```



School		Student		
id	name	id	name	school_id
1	Some School	1	Paul	2
2	Another School	2	William	NULL
		3	John	NULL
		4	Steven	1
		5	George	1



Referential integrity ensures **foreign keys always reference valid primary keys**.

Terminology

- **Referenced Relation** → Contains Primary Key
- **Referencing Relation** → Contains Foreign Key

Operational Constraints

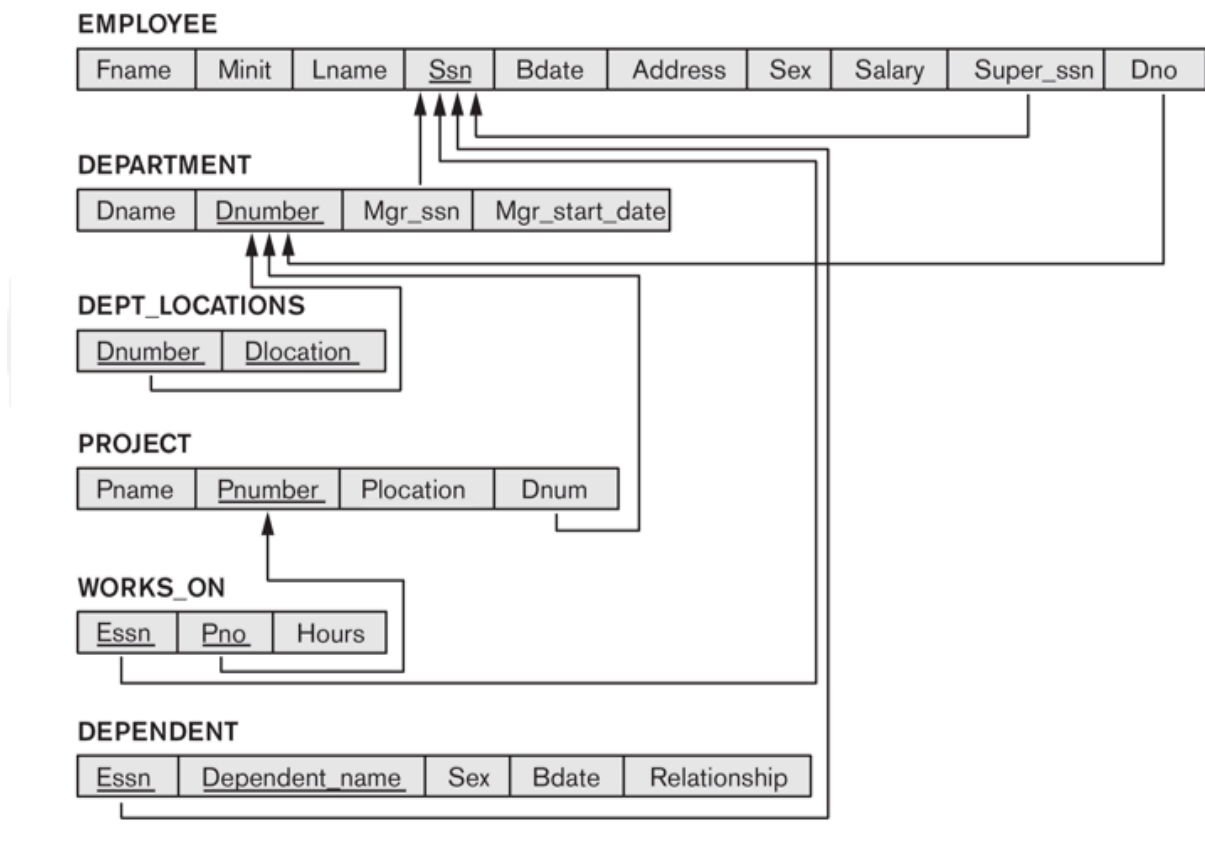
Operation	Referenced (PK)	Referencing (FK)
Insert	Allowed	Restricted
Delete	Restricted	Allowed
Update	Common	Rare

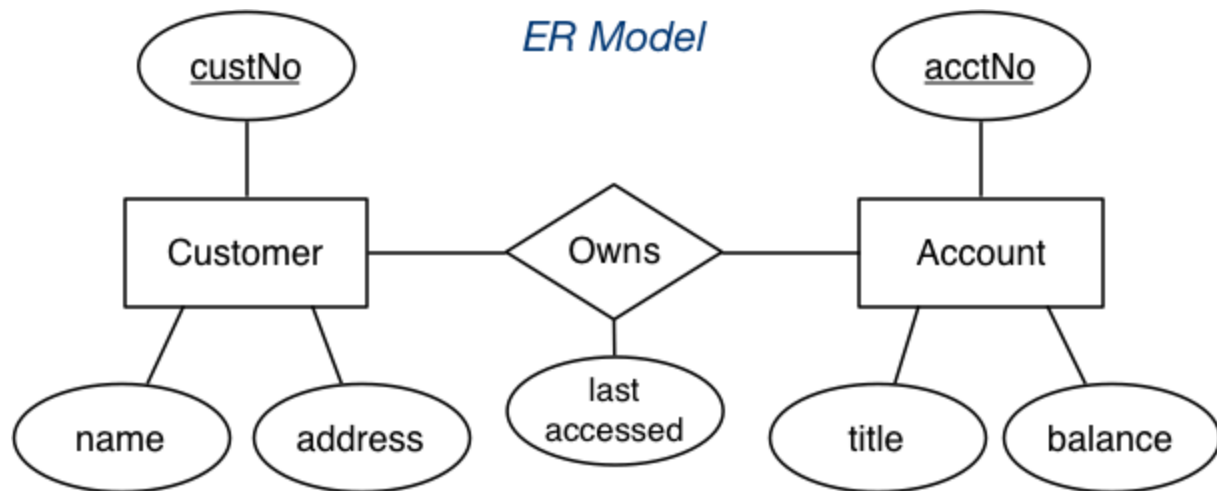
Deletion Handling Strategies

1. **NO ACTION**
 - Reject deletion
2. **SET NULL**
 - FK set to NULL
3. **CASCADE**
 - Deletes dependent rows automatically ⚠

CASCADE is powerful but dangerous in large systems.

6. Converting ER Models to RDBMS





Relational Version

Customer	custNo	name	address
----------	---------------	------	---------

Account	acctNo	title	balance
---------	---------------	-------	---------

Owns	acctNo	custNo	lastAccessed
------	---------------	---------------	--------------

Mapping Rules

1. **Strong Entity → Table**
 - Attributes → Columns
 - Key → Primary Key
2. **Composite Attributes**
 - Flatten into atomic columns
3. **Multi-valued Attributes**
 - Create **separate table**
 - PK = (Original PK + Multivalued attribute)
4. **Weak Entity**
 - Include PK of strong entity
 - Composite primary key

ER conversion ensures **1NF compliance**.